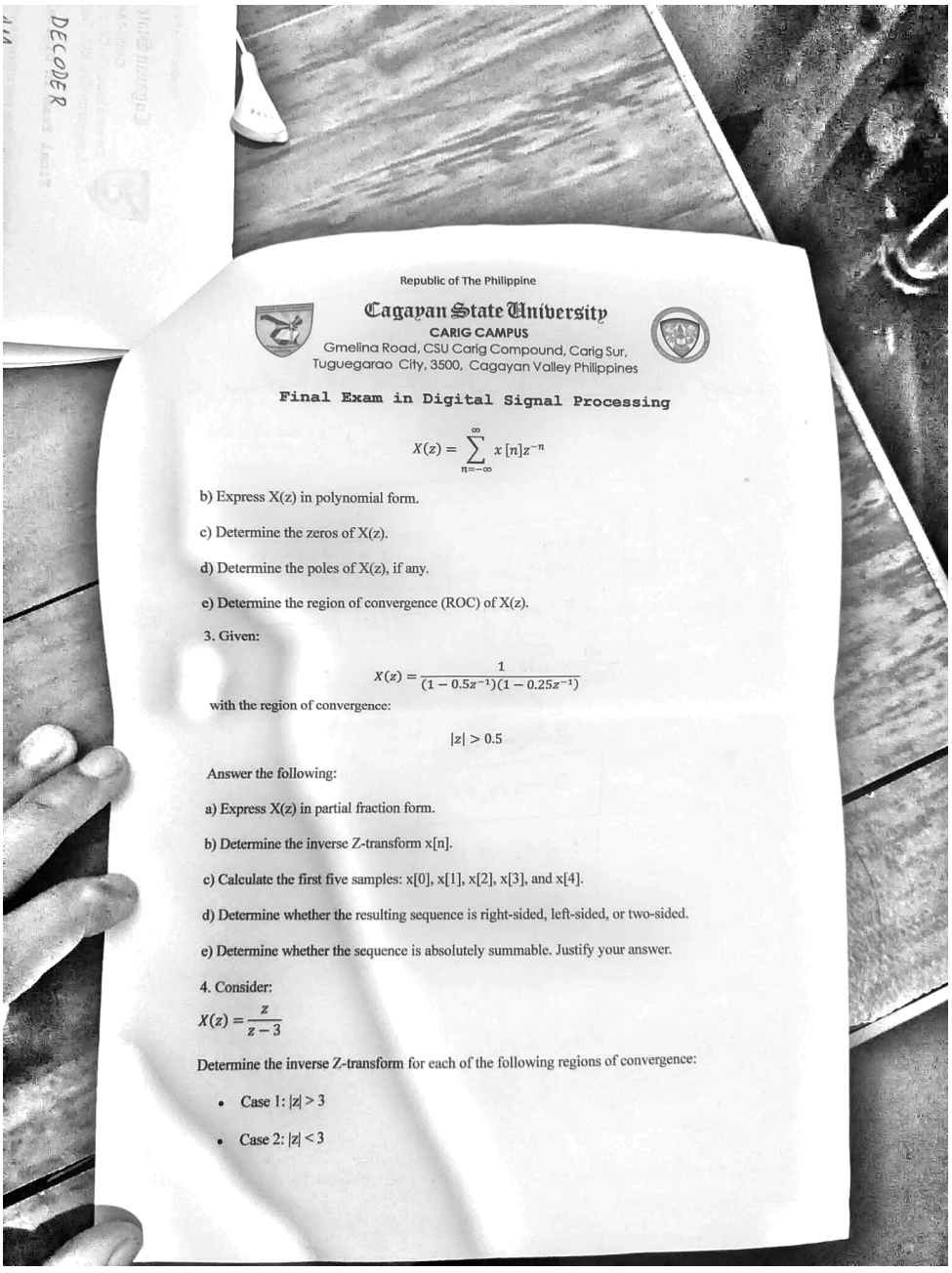
SmartBoard Session Report

Course: Cpe13 - Embedded
Session: designing
Date: 2026-10-24
Status: draft



Cleaned image (gentle)

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vag yl. Exam in Digital Signal Processing aS san fel
chk 3 X@= D) xf ae
eae ey A b) Express X(z) in polynomial form. ro 2 “ Lae
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ee c) Determine the zeros of X(z). we °
BT Ss eT oa "1
BBs d) Determine the poles of X(z), if any. era
ben aa ¢) Determine the region of convergence (ROC) of X(z). cae one
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re 3. Given: a By
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Megat with the region of convergence: LA at a
A Ny Answer the following: ee
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Pm st a) Express X(z) in partial fraction form. captor tit Page
nr Aare b) Determine the inverse Z-transform x[n]. ae er %
ae ~ ao c) Calculate the first five samples: x[0], x{1], x{2}, x[3}, and x[4]. Ny * 5 ae
ar at Re d) Determine whether the resulting sequence is right-sided, left-sided, or
two-sided. - Aoce =
ie A ¢) Determine whether the sequence is absolutely summable, Justify your answer. at We.
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=" Lx, oe 4. Consider: ye ORY cea
at gee Determine the inverse Z-transform for each of the following regions of convergence:
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